FEA HW3

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1.1

Before applying the Galerkin method we define the residuals in the domain and on the non-Dirichlet boundaries, introducing an approximate form of the solution $\tilde{u} = \phi_j a_j$:

$$R = -\frac{d}{dx}\left[EA_0(1+\frac{x}{L})\frac{du}{dx}\right] - f(x) = -\frac{d}{dx}\left[EA_0(1+\frac{x}{L})\frac{d\phi_j}{dx}\right]a_j - f(x)$$

$$r_1 = \left[-EA_0(1+\frac{x}{L})\frac{du}{dx} + ku\right]_{x=0} = \left[-EA_0(1+\frac{x}{L})\frac{d\phi_j}{dx} + k\phi_j\right]a_j\Big|_{x=0}$$

$$r_2 = \left[EA_0(1+\frac{x}{L})\frac{du}{dx}\right]_{x=L} - F = \left[EA_0(1+\frac{x}{L})\frac{d\phi_j}{dx}\right]a_j\Big|_{x=L} - F$$

Now we demand

$$\int_0^L \phi_i R dx + \phi_i(0)r_1 + \phi_i(L)r_2 = 0$$

$$\Rightarrow \int_0^L \phi_i\Bigg(-\frac{d}{dx}\left[EA_0(1+\frac{x}{L})\frac{d\phi_j}{dx}\right]a_j - f(x)\Bigg)dx + \phi_i(0)\Bigg(\left[-EA_0(1+\frac{x}{L})\frac{d\phi_j}{dx} + k\phi_j\right]a_j\Bigg)_{x=0}$$

$$+\phi_i(L)\Bigg(\left[EA_0(1+\frac{x}{L})\frac{d\phi_j}{dx}\right]a_j\Big|_{x=L} - F\Bigg) = 0$$

$$\Rightarrow \int_0^L \phi_i\Bigg(-\frac{d}{dx}\left[EA_0(1+\frac{x}{L})\frac{d\phi_j}{dx}\right]a_j\Bigg)dx + (k\phi_i\phi_j a_j)_{x=0} + \phi_i\left[EA_0(1+\frac{x}{L})\frac{d\phi_j}{dx}\right]a_j\Bigg|_{x=0}^{x=L} = \int_0^L \phi_i f(x)dx + \phi_i(L)F$$

$$\phi_i\Bigg(-\frac{d}{dx}\left[EA_0(1+\frac{x}{L})\frac{d\phi_j}{dx}\right]a_j\Bigg) = \frac{d\phi_i}{dx}\Bigg(EA_0(1+\frac{x}{L})\frac{d\phi_j}{dx}\Bigg)a_j - \frac{d}{dx}\Bigg(\phi_i EA_0(1+\frac{x}{L})\frac{d\phi_j}{dx}a_j\Bigg)$$

$$\Rightarrow \int_0^L \frac{d\phi_i}{dx}\Bigg(EA_0(1+\frac{x}{L})\frac{d\phi_j}{dx}\Bigg)a_j dx - \Bigg(\phi_i EA_0(1+\frac{x}{L})\frac{d\phi_j}{dx}a_j\Bigg)_{x=0}^{x=L} + (k\phi_i\phi_j a_j)_{x=0} + \phi_i\left[EA_0(1+\frac{x}{L})\frac{d\phi_j}{dx}\right]a_j\Bigg|_{x=0}^{x=L}$$

$$= \int_0^L \phi_i f(x)dx + \phi_i(L)F$$

$$\Rightarrow \int_0^L \frac{d\phi_i}{dx}\Bigg(EA_0(1+\frac{x}{L})\frac{d\phi_j}{dx}\Bigg)a_j dx + (k\phi_i\phi_j a_j)_{x=0} = \int_0^L \phi_i f(x)dx + \phi_i(L)F$$

Neumann boundary terms have cancelled out which is the expected result. Due to first BC being mixed, the k term in the equation represents the Dirichlet term in the first BC, and there is no need to constrain the shape functions any further at this time. This is the weak formulation of the problem and we will write it as

$$(K_{ij}^{\kappa} + \bar{K}_{ij})a_j = \int_0^L \phi_i f(x)dx + \bar{F}_i$$

1.2

To solve the problem with $n = 6$ linear elements we introduce $N = n + 1 = 7$ nodal points x_j, and the following transformation $\forall e \leq n \in \mathbb{N}$ (e is not in Einstein notation):

$$\xi^e(x) = \frac{2}{h^e}x - \frac{x_e + x_{e+1}}{h^e};\ x(\xi^e) = \frac{h^e}{2}\xi^e + \frac{x_e + x_{e+1}}{2} = \frac{L}{2n}\xi^e + \frac{eL}{n} - \frac{L}{2n} = \frac{L}{12}(\xi^e - 1 + 2e)$$

Where

$$h^e = \frac{L}{n} = \frac{L}{6}$$

$$x_j = (j-1)h^e = (j-1)\frac{L}{n} = (j-1)\frac{L}{6}$$

$$J^e = \frac{dx}{d\xi^e} = \frac{h^e}{2} = \frac{L}{12}$$

For each element there are only two shape functions so we will define

$$\begin{cases} \phi_1^e(x) = \phi_e(x) \\ \phi_2^e(x) = \phi_{e+1}(x) \end{cases}$$

$$\begin{cases} \hat{\phi}_1(\xi^e) = \frac{1}{2}(1 - \xi^e) \\ \hat{\phi}_2(\xi^e) = \frac{1}{2}(1 + \xi^e) \end{cases}$$

$$\begin{cases} B_1(\xi^e) = \frac{d\hat{\phi}_1(\xi^e)}{d\xi^e} = -\frac{1}{2} \\ B_2(\xi^e) = \frac{d\hat{\phi}_2(\xi^e)}{d\xi^e} = \frac{1}{2} \end{cases}$$

Note that

$$\phi_1^e(x(\xi^e)) = \hat{\phi}_1(\xi^e)$$

$$\phi_2^e(x(\xi^e)) = \hat{\phi}_2(\xi^e)$$

Therefore

$$\left.\frac{d\phi_i^e(x)}{dx}\right|_{x(\xi^e)} = \frac{d\phi_i^e(x(\xi^e))}{d\xi^e}\frac{d\xi^e}{dx} = \frac{d\hat{\phi}_i(\xi^e)}{d\xi^e}\frac{1}{J^e} = B_i(\xi^e)\frac{2}{h^e}$$

The integral over the domain can be split into a sum of integrals over each element, and since the only nonzero shape functions over an element are its own, we can say

$$K_{ij}^{\kappa} = \int_0^L \frac{d\phi_i}{dx}\left(EA_0(1 + \frac{x}{L})\frac{d\phi_j}{dx}\right)dx$$

$$K_{ij}^{e} = \int_{x_e}^{x_{e+1}} \frac{d\phi_i^e}{dx}\left(EA_0(1 + \frac{x}{L})\frac{d\phi_j^e}{dx}\right)dx$$

$$\mathbf{K}^{\kappa} = \mathop{A}_{e=1}^{n} \mathbf{K^e}$$

Where A is the special matrix summation specific in tutorial 3 of the course. To finish writing the integral form of the stiffness matrix for a master element we simplify to

$$K_{ij}^e = \int_{-1}^{1} \frac{1}{J^e}B_i EA_0(1 + \frac{x(\xi^e)}{L})\frac{1}{J^e}B_j J^e d\xi^e = \frac{2}{h^e}\int_{-1}^{1} B_i EA_0 \frac{1}{2n}(\xi^e + 2e + 2n - 1)B_j d\xi^e$$

$$= \frac{2}{h^e}\int_{-1}^{1} B_i EA_0 \frac{1}{12}(\xi^e + 2e + 11)B_j d\xi^e$$

We would now like to calculate the above matrix explicitly using Gaussian quadrature. Since our integrand is a linear function we only need $2p - 1 \geq 1 \Rightarrow \mathbb{N} \ni p \geq 1$ Gauss point and our integral value will be exact.

$$\xi_k^e = 0;\ w_k = 2$$

$$K^e_{ij} = \sum_{k=1}^{1} w_k \frac{2}{h^e} B_i E A_0 \frac{1}{2n}(2n+2e-1)B_j$$
$$= \sum_{k=1}^{1} w_k \frac{2}{h^e} B_i E A_0 \frac{1}{12}(11+2e)B_j$$
$$\Rightarrow \mathbf{K^e} = \frac{EA_0(2n+2e-1)}{2L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

To calculate the integral form of the forcing vector we introduce the following approximation without introducing additional error:

$$f(x) \approx f(x_j)\phi_j = q(L+x_j)\phi_j$$

This allows us to write the forcing term as

$$\left[\int_0^L \phi_i \phi_j dx\right] q(L+x_j)$$

This allows us to say once again:

$$M^g_{ij} = \int_0^L \phi_i \phi_j dx$$
$$M^e_{ij} = \int_{x_e}^{x_{e+1}} \phi^e_i \phi^e_j dx$$
$$\mathbf{M^g} = \mathop{A}_{e=1}^{n} \mathbf{M^e}$$
$$M^e_{ij} = \int_{-1}^{1} \hat{\phi}_i(\xi^e)\hat{\phi}_j(\xi^e) J^e d\xi^e = \frac{h^e}{2}\int_{-1}^{1} \hat{\phi}_i(\xi^e)\hat{\phi}_j(\xi^e) d\xi^e$$

Once again for Gaussian quadrature we need $2p-1 \geq 2 \Rightarrow \mathbb{N} \ni p \geq 2$ Gauss points since our integrand is quadratic:

$$\xi^e_1 = \frac{1}{\sqrt{3}};\ \xi^e_2 = -\frac{1}{\sqrt{3}};\ w_k = 1$$
$$\frac{2}{h^e} M^e_{ij} = \sum_{k=1}^{2} w_k \hat{\phi}_i(\xi^e_k)\hat{\phi}_i(\xi^e_k)$$
$$\Rightarrow \mathbf{M^e} = \frac{h^e}{6}\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Calculation of the mass matrix can be found in a previous homework or in the tutorials. It should be noted that the obtaining the mass matrix with single point Gauss quadrature will yield an inexact integral but will not introduce additional error. The global forcing function is simply $f_j = q(L+x_j)$ so the local forcing function is

$$\mathbf{f^e} = q\begin{bmatrix} L+x_e \\ L+x_{e+1} \end{bmatrix}$$

So the local forcing vector is

$$\mathbf{F^e} = \mathbf{M^e f^e} = q\frac{h^e}{6}\begin{bmatrix} 2(L+x_e)+L+x_{e+1} \\ L+x_e+2(L+x_{e+1}) \end{bmatrix} = \frac{qL}{6n}\begin{bmatrix} 3L+2x_e+x_{e+1} \\ 3L+x_e+2x_{e+1} \end{bmatrix}$$

The boundary terms can easily be calculated as

$$\bar{K}_{ij} = \begin{cases} k & i=j=1 \\ 0 & otherwise \end{cases}$$

$$\bar{F}_i = \begin{cases} F & i=L \\ 0 & otherwise \end{cases}$$

Putting it all together we have

$$\bar{\mathbf{K}} = \begin{bmatrix} k & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} ; \ \bar{\mathbf{F}} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ F \end{bmatrix}$$

$$\mathbf{K^e} = \frac{11EA_0}{2L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} + \frac{EA_0}{2L}\begin{bmatrix} 2e & -2e \\ -2e & 2e \end{bmatrix}$$

$$\Rightarrow \mathbf{K^g} = \frac{EA_0}{2L}\begin{bmatrix} 11 & -11 & 0 & 0 & 0 & 0 & 0 \\ -11 & 22 & -11 & 0 & 0 & 0 & 0 \\ 0 & -11 & 22 & -11 & 0 & 0 & 0 \\ 0 & 0 & -11 & 22 & -11 & 0 & 0 \\ 0 & 0 & 0 & -11 & 22 & -11 & 0 \\ 0 & 0 & 0 & 0 & -11 & 22 & -11 \\ 0 & 0 & 0 & 0 & 0 & -11 & 11 \end{bmatrix} + \frac{EA_0}{2L}\begin{bmatrix} 2 & -2 & 0 & 0 & 0 & 0 & 0 \\ -2 & 6 & -4 & 0 & 0 & 0 & 0 \\ 0 & -4 & 10 & -6 & 0 & 0 & 0 \\ 0 & 0 & -6 & 14 & -8 & 0 & 0 \\ 0 & 0 & 0 & -8 & 18 & -10 & 0 \\ 0 & 0 & 0 & 0 & -10 & 22 & -12 \\ 0 & 0 & 0 & 0 & 0 & -12 & 12 \end{bmatrix}$$

$$= \frac{EA_0}{2L}\begin{bmatrix} 13 & -13 & 0 & 0 & 0 & 0 & 0 \\ -13 & 28 & -15 & 0 & 0 & 0 & 0 \\ 0 & -15 & 32 & -17 & 0 & 0 & 0 \\ 0 & 0 & -17 & 36 & -19 & 0 & 0 \\ 0 & 0 & 0 & -19 & 40 & -21 & 0 \\ 0 & 0 & 0 & 0 & -21 & 44 & -23 \\ 0 & 0 & 0 & 0 & 0 & -23 & 23 \end{bmatrix}$$

$$\mathbf{F^g} = \frac{qL}{36}\begin{bmatrix} 3L + 2x_1 + x_2 \\ 6L + x_1 + 4x_2 + x_3 \\ 6L + x_2 + 4x_3 + x_4 \\ 6L + x_3 + 4x_4 + x_5 \\ 6L + x_4 + 4x_5 + x_6 \\ 6L + x_5 + 4x_6 + x_7 \\ 3L + x_6 + 2x_7 \end{bmatrix} = \frac{qL^2}{36}\begin{bmatrix} \frac{19}{6} \\ 7 \\ 8 \\ 9 \\ 10 \\ 11 \\ \frac{35}{6} \end{bmatrix}$$

$$(\mathbf{K}^\kappa + \bar{\mathbf{K}})\mathbf{a} = \mathbf{F^g} + \bar{\mathbf{F}} \Rightarrow$$

$$\frac{EA_0}{2L}\begin{bmatrix} 13 + \frac{2Lk}{EA_0} & -13 & 0 & 0 & 0 & 0 & 0 \\ -13 & 28 & -15 & 0 & 0 & 0 & 0 \\ 0 & -15 & 32 & -17 & 0 & 0 & 0 \\ 0 & 0 & -17 & 36 & -19 & 0 & 0 \\ 0 & 0 & 0 & -19 & 40 & -21 & 0 \\ 0 & 0 & 0 & 0 & -21 & 44 & -23 \\ 0 & 0 & 0 & 0 & 0 & -23 & 23 \end{bmatrix}\begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \\ a_6 \\ a_7 \end{bmatrix} = \frac{qL^2}{36}\begin{bmatrix} \frac{19}{6} \\ 7 \\ 8 \\ 9 \\ 10 \\ 11 \\ \frac{35}{6} + \frac{36F}{qL^2} \end{bmatrix}$$

Plugging in values and using MATLAB matrix inversion,

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \\ a_6 \\ a_7 \end{bmatrix} = \begin{bmatrix} 0.2500 \\ 0.2537 \\ 0.2567 \\ 0.2590 \\ 0.2609 \\ 0.2623 \\ 0.2633 \end{bmatrix}$$

Finally, our choice of shape functions allows us to make the simplification

$$\tilde{u} = \phi_j a_j \Rightarrow \tilde{u} = \begin{cases} \phi_j^e a_{e+j-1} & x_e \le x \le x_{e+1} \end{cases}$$

Again, this applies $\forall e$; e is not in Einstein notation.

$$\phi_1^1(x) = \hat{\phi}_1(\xi^1(x)) = \frac{1}{2}\left(1 - (\frac{2}{h^e}x - \frac{x_1 + x_2}{h^e})\right) = 1 - 6x$$

$$\phi_2^1(x) = \hat{\phi}_2(\xi^1(x)) = \frac{1}{2}\left(1 + (\frac{2}{h^e}x - \frac{x_1 + x_2}{h^e})\right) = 6x$$

$$\phi_1^2(x) = \hat{\phi}_1(\xi^2(x)) = \frac{1}{2}\left(1 - (\frac{2}{h^e}x - \frac{x_2 + x_3}{h^e})\right) = 2 - 6x$$

$$\phi_2^2(x) = \hat{\phi}_2(\xi^2(x)) = \frac{1}{2}\left(1 + (\frac{2}{h^e}x - \frac{x_2 + x_3}{h^e})\right) = 6x - 1$$

$$\phi_1^3(x) = \hat{\phi}_1(\xi^3(x)) = \frac{1}{2}\left(1 - (\frac{2}{h^e}x - \frac{x_3 + x_4}{h^e})\right) = 3 - 6x$$

$$\phi_2^3(x) = \hat{\phi}_2(\xi^3(x)) = \frac{1}{2}\left(1 + (\frac{2}{h^e}x - \frac{x_3 + x_4}{h^e})\right) = 6x - 2$$

$$\phi_1^4(x) = \hat{\phi}_1(\xi^4(x)) = \frac{1}{2}\left(1 - (\frac{2}{h^e}x - \frac{x_4 + x_5}{h^e})\right) = 4 - 6x$$

$$\phi_2^4(x) = \hat{\phi}_2(\xi^4(x)) = \frac{1}{2}\left(1 + (\frac{2}{h^e}x - \frac{x_4 + x_5}{h^e})\right) = 6x - 3$$

$$\phi_1^5(x) = \hat{\phi}_1(\xi^5(x)) = \frac{1}{2}\left(1 - (\frac{2}{h^e}x - \frac{x_5 + x_6}{h^e})\right) = 5 - 6x$$

$$\phi_2^5(x) = \hat{\phi}_2(\xi^5(x)) = \frac{1}{2}\left(1 + (\frac{2}{h^e}x - \frac{x_5 + x_6}{h^e})\right) = 6x - 4$$

$$\phi_1^6(x) = \hat{\phi}_1(\xi^6(x)) = \frac{1}{2}\left(1 - (\frac{2}{h^e}x - \frac{x_6 + x_7}{h^e})\right) = 6 - 6x$$

$$\phi_2^6(x) = \hat{\phi}_2(\xi^6(x)) = \frac{1}{2}\left(1 + (\frac{2}{h^e}x - \frac{x_6 + x_7}{h^e})\right) = 6x - 5$$

$$\Rightarrow \tilde{u} = \begin{cases} 0.2500(1-6x) + 0.2537(6x) & 0 \leq x \leq \frac{1}{6} \\ 0.2537(2-6x) + 0.2567(6x-1) & \frac{1}{6} \leq x \leq \frac{2}{6} \\ 0.2567(3-6x) + 0.2590(6x-2) & \frac{2}{6} \leq x \leq \frac{3}{6} \\ 0.2590(4-6x) + 0.2609(6x-3) & \frac{3}{6} \leq x \leq \frac{4}{6} \\ 0.2609(5-6x) + 0.2623(6x-4) & \frac{4}{6} \leq x \leq \frac{5}{6} \\ 0.2623(6-6x) + 0.2633(6x-5) & \frac{5}{6} \leq x \leq 1 \end{cases}$$

To find the flux we differentiate the solution:

$$\tilde{q} = \frac{d\tilde{u}}{dx} = \begin{cases} \frac{d\phi_j^e(x)}{dx} a_{e+j-1} & x_e \leq x \leq x_{e+1} \end{cases}$$

$$= \begin{cases} B_j(\xi^e(x)) \frac{2}{h^e} a_{e+j-1} & x_e \leq x \leq x_{e+1} \end{cases}$$

$$= \begin{cases} 6(-1)^j a_{e+j-1} & x_e \leq x \leq x_{e+1} \end{cases}$$

$$\Rightarrow \tilde{q} = \begin{cases} 0.2500(-6) + 0.2537(6) & 0 \leq x \leq \frac{1}{6} \\ 0.2537(-6) + 0.2567(6) & \frac{1}{6} \leq x \leq \frac{2}{6} \\ 0.2567(-6) + 0.2590(6) & \frac{2}{6} \leq x \leq \frac{3}{6} \\ 0.2590(-6) + 0.2609(6) & \frac{3}{6} \leq x \leq \frac{4}{6} \\ 0.2609(-6) + 0.2623(6) & \frac{4}{6} \leq x \leq \frac{5}{6} \\ 0.2623(-6) + 0.2633(6) & \frac{5}{6} \leq x \leq 1 \end{cases}$$

1.3

To solve the problem with $n = 3$ elements of order $r = 2$ we introduce the same $N = 2n + 1 = 7$ nodal points x_j, and generalize the previous transformation $\forall e \leq n \in \mathbb{N}$ to any order (assuming equally spaced nodes):

$$x(\xi^e) = \frac{h^e}{2}\xi^e + \frac{x_{er+1-r} + x_{er+1}}{2} = \frac{L}{2n}\xi^e + \frac{eL}{n} - \frac{L}{2n} = \frac{L}{2n}\Big(\xi^e - 1 + 2e\Big)$$

Where

$$h^e = \frac{L}{n}$$

$$x_j = (j-1)\frac{L}{rn}$$

$$J^e = \frac{dx}{d\xi^e} = \frac{h^e}{2}$$

$$\Big\{\phi_i^e(x) = \phi_{er+i-r}(x) \quad r+1 \geq i \in \mathbb{N}$$

As we can see, the overall transformation is independent of element order which is expected. Expressions for $\hat{\phi}_i(\xi^e);\ r+1 \geq i \in \mathbb{N}$ are derived in tutorials for each r using the properties of local support and completeness.

$$\Big\{\hat{B}_i(\xi^e) = \tfrac{d\hat{\phi}_i(\xi^e)}{d\xi^e} \quad r+1 \geq i \in \mathbb{N}$$

So, $\forall r \in \mathbb{N}$ we have:

$$\phi_i^e(x(\xi^e)) = \hat{\phi}_i(\xi^e)$$

$$\Rightarrow \left.\frac{d\phi_i^e(x)}{dx}\right|_{x(\xi^e)} = \frac{d\phi_i^e(x(\xi^e))}{d\xi^e}\frac{d\xi^e}{dx} = \frac{d\hat{\phi}_i(\xi^e)}{d\xi^e}\frac{1}{J^e} = \hat{B}_i(\xi^e)\frac{2}{h^e}$$

$$K_{ij}^{\kappa} = \int_0^L \frac{d\phi_i}{dx}\Big(EA_0(1+\frac{x}{L})\frac{d\phi_j}{dx}\Big)dx$$

$$K_{ij}^{e} = \int_{x_{er-r+1}}^{x_{er+1}} \frac{d\phi_i^e}{dx}\Big(EA_0(1+\frac{x}{L})\frac{d\phi_j^e}{dx}\Big)dx$$

$$\mathbf{K}^{\kappa} = \mathop{A}_{e=1}^{n} \mathbf{K^e}$$

We can now simplify:

$$K_{ij}^e = \int_{-1}^{1} \frac{1}{J^e}\hat{B}_i EA_0(1+\frac{x(\xi^e)}{L})\frac{1}{J^e}\hat{B}_j J^e d\xi^e = \frac{2}{h^e}\int_{-1}^{1}\hat{B}_i EA_0 \frac{1}{2n}\Big(\xi^e - 1 + 2n + 2e\Big)\hat{B}_j d\xi^e$$

At this point we need to introduce the shape functions for quadratic elements. For $r = 2$:

$$\begin{cases} \hat{\phi}_1(\xi^e) = -\frac{1}{2}\xi^e(1-\xi^e) \\ \hat{\phi}_2(\xi^e) = (1+\xi^e)(1-\xi^e) \\ \hat{\phi}_3(\xi^e) = \frac{1}{2}\xi^e(1+\xi^e) \end{cases}$$

$$\begin{cases} \hat{B}_1(\xi^e) = \frac{d\hat{\phi}_1(\xi^e)}{d\xi^e} = \xi^e - \frac{1}{2} \\ \hat{B}_2(\xi^e) = \frac{d\hat{\phi}_2(\xi^e)}{d\xi^e} = -2\xi^e \\ \hat{B}_3(\xi^e) = \frac{d\hat{\phi}_3(\xi^e)}{d\xi^e} = \xi^e + \frac{1}{2} \end{cases}$$

This implies that our integrand is cubic and therefore we need $2p - 1 \geq 3 \Rightarrow \mathbb{N} \ni p \geq 2$ Gauss points for exact integral value.

$$\xi_1^e = \frac{1}{\sqrt{3}};\ \xi_2^e = -\frac{1}{\sqrt{3}};\ w_k = 1$$

$$K^e_{ij} = \sum_{k=1}^{2} w_k \hat{B}_i EA_0 \frac{1}{h^e n}(\xi^e + 2n + 2e - 1)\hat{B}_j$$

$$K^e_{11} = \frac{EA_0}{h^e n}[\hat{B}_1(\frac{1}{\sqrt{3}})\hat{B}_1(\frac{1}{\sqrt{3}})(\frac{1}{\sqrt{3}} + 2n + 2e - 1) + \hat{B}_1(-\frac{1}{\sqrt{3}})\hat{B}_1(-\frac{1}{\sqrt{3}})(-\frac{1}{\sqrt{3}} + 2n + 2e - 1)]$$

$$K^e_{12} = K^e_{21} = \frac{EA_0}{h^e n}[\hat{B}_1(\frac{1}{\sqrt{3}})\hat{B}_2(\frac{1}{\sqrt{3}})(\frac{1}{\sqrt{3}} + 2n + 2e - 1) + \hat{B}_1(-\frac{1}{\sqrt{3}})\hat{B}_2(-\frac{1}{\sqrt{3}})(-\frac{1}{\sqrt{3}} + 2n + 2e - 1)]$$

$$K^e_{13} = K^e_{31} = \frac{EA_0}{h^e n}[\hat{B}_1(\frac{1}{\sqrt{3}})\hat{B}_3(\frac{1}{\sqrt{3}})(\frac{1}{\sqrt{3}} + 2n + 2e - 1) + \hat{B}_1(-\frac{1}{\sqrt{3}})\hat{B}_3(-\frac{1}{\sqrt{3}})(-\frac{1}{\sqrt{3}} + 2n + 2e - 1)]$$

$$K^e_{22} = \frac{EA_0}{h^e n}[\hat{B}_2(\frac{1}{\sqrt{3}})\hat{B}_2(\frac{1}{\sqrt{3}})(\frac{1}{\sqrt{3}} + 2n + 2e - 1) + \hat{B}_2(-\frac{1}{\sqrt{3}})\hat{B}_2(-\frac{1}{\sqrt{3}})(-\frac{1}{\sqrt{3}} + 2n + 2e - 1)]$$

$$K^e_{23} = K^e_{32} = \frac{EA_0}{h^e n}[\hat{B}_2(\frac{1}{\sqrt{3}})\hat{B}_3(\frac{1}{\sqrt{3}})(\frac{1}{\sqrt{3}} + 2n + 2e - 1) + \hat{B}_2(-\frac{1}{\sqrt{3}})\hat{B}_3(-\frac{1}{\sqrt{3}})(-\frac{1}{\sqrt{3}} + 2n + 2e - 1)]$$

$$K^e_{33} = \frac{EA_0}{h^e n}[\hat{B}_3(\frac{1}{\sqrt{3}})\hat{B}_3(\frac{1}{\sqrt{3}})(\frac{1}{\sqrt{3}} + 2n + 2e - 1) + \hat{B}_3(-\frac{1}{\sqrt{3}})\hat{B}_3(-\frac{1}{\sqrt{3}})(-\frac{1}{\sqrt{3}} + 2n + 2e - 1)]$$

So

$$K^e_{11} = \frac{EA_0}{h^e n}[(\frac{1}{\sqrt{3}} - \frac{1}{2})(\frac{1}{\sqrt{3}} - \frac{1}{2})(\frac{1}{\sqrt{3}} + 2n + 2e - 1) + (-\frac{1}{\sqrt{3}} - \frac{1}{2})(-\frac{1}{\sqrt{3}} - \frac{1}{2})(-\frac{1}{\sqrt{3}} + 2n + 2e - 1)]$$

$$K^e_{12} = K^e_{21} = \frac{EA_0}{h^e n}[(\frac{1}{\sqrt{3}} - \frac{1}{2})(-2\frac{1}{\sqrt{3}})(\frac{1}{\sqrt{3}} + 2n + 2e - 1) + (-\frac{1}{\sqrt{3}} - \frac{1}{2})(2\frac{1}{\sqrt{3}})(-\frac{1}{\sqrt{3}} + 2n + 2e - 1)]$$

$$K^e_{13} = K^e_{31} = \frac{EA_0}{h^e n}[(\frac{1}{\sqrt{3}} - \frac{1}{2})(\frac{1}{\sqrt{3}} + \frac{1}{2})(\frac{1}{\sqrt{3}} + 2n + 2e - 1) + (-\frac{1}{\sqrt{3}} - \frac{1}{2})(-\frac{1}{\sqrt{3}} + \frac{1}{2})(-\frac{1}{\sqrt{3}} + 2n + 2e - 1)]$$

$$K^e_{22} = \frac{EA_0}{h^e n}[(-2\frac{1}{\sqrt{3}})(-2\frac{1}{\sqrt{3}})(\frac{1}{\sqrt{3}} + 2n + 2e - 1) + (2\frac{1}{\sqrt{3}})(2\frac{1}{\sqrt{3}})(-\frac{1}{\sqrt{3}} + 2n + 2e - 1)]$$

$$K^e_{23} = K^e_{32} = \frac{EA_0}{h^e n}[(-2\frac{1}{\sqrt{3}})(\frac{1}{\sqrt{3}} + \frac{1}{2})(\frac{1}{\sqrt{3}} + 2n + 2e - 1) + (2\frac{1}{\sqrt{3}})(-\frac{1}{\sqrt{3}} + \frac{1}{2})(-\frac{1}{\sqrt{3}} + 2n + 2e - 1)]$$

$$K^e_{33} = \frac{EA_0}{h^e n}[(\frac{1}{\sqrt{3}} + \frac{1}{2})(\frac{1}{\sqrt{3}} + \frac{1}{2})(\frac{1}{\sqrt{3}} + 2n + 2e - 1) + (-\frac{1}{\sqrt{3}} + \frac{1}{2})(-\frac{1}{\sqrt{3}} + \frac{1}{2})(-\frac{1}{\sqrt{3}} + 2n + 2e - 1)]$$

So

$$K^e_{11} = \frac{EA_0}{h^e n}(\frac{7}{3}n - \frac{2}{3} - \frac{7}{6} + \frac{7}{3}e)$$

$$K^e_{12} = K^e_{21} = \frac{EA_0}{h^e n}(-\frac{8}{3}n + \frac{2}{3} + \frac{8}{6} - \frac{8}{3}e)$$

$$K^e_{13} = K^e_{31} = \frac{EA_0}{h^e n}(\frac{1}{3}n - \frac{1}{6} + \frac{1}{3}e)$$

$$K^e_{22} = \frac{EA_0}{h^e n}(\frac{16}{3}n - \frac{16}{6} + \frac{16}{3}e)$$

$$K^e_{23} = K^e_{32} = \frac{EA_0}{h^e n}(-\frac{8}{3}n - \frac{2}{3} - \frac{8}{6} - \frac{8}{3}e)$$

$$K^e_{33} = \frac{EA_0}{h^e n}(\frac{7}{3}n + \frac{2}{3} - \frac{7}{6} + \frac{7}{3}e)$$

$$\Rightarrow \mathbf{K^e} = \frac{EA_0}{6L}\left((2n + 2e - 1)\begin{bmatrix} 7 & -8 & 1 \\ -8 & 16 & -8 \\ 1 & -8 & 7 \end{bmatrix} + 4\begin{bmatrix} -1 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & 1 \end{bmatrix}\right)$$

Considering that the first matrix term in the above expression was shown to be correct in tutorials, this result is not unexpected. In our case $n = 3$ so

$$\mathbf{K^e} = \frac{EA_0}{12L}\left(\begin{bmatrix} 62 & -72 & 10 \\ -72 & 160 & -88 \\ 10 & -88 & 78 \end{bmatrix} + 4e\begin{bmatrix} 7 & -8 & 1 \\ -8 & 16 & -8 \\ 1 & -8 & 7 \end{bmatrix}\right)$$

To calculate the integral form of the forcing vector we once again approximate $f(x)$ and calculate the mass matrix:

$$f(x) \approx f(x_j)\phi_j = q(L + x_j)\phi_j$$

$$\left[\int_0^L \phi_i \phi_j dx\right] q(L + x_j)$$

$$M_{ij}^g = \int_0^L \phi_i \phi_j dx$$

$$M_{ij}^e = \int_{x_{er+1-r}}^{x_{er+1}} \phi_i^e \phi_j^e dx$$

$$\mathbf{M^g} = \mathop{A}_{e=1}^{n} \mathbf{M^e}$$

$$M_{ij}^e = \int_{-1}^{1} \hat{\phi}_i(\xi^e)\hat{\phi}_j(\xi^e) J^e d\xi^e = \frac{h^e}{2}\int_{-1}^{1} \hat{\phi}_i(\xi^e)\hat{\phi}_j(\xi^e) d\xi^e$$

Given $r = 2$ we need $2p - 1 \geq 4 \Rightarrow \mathbb{N} \ni p \geq 3$ Gauss points for Gaussian quadrature. This mass matrix is calculated in a previous tutorial:

$$\mathbf{M^e} = \frac{h^e}{30}\begin{bmatrix} 4 & 2 & -1 \\ 2 & 16 & 2 \\ -1 & 2 & 4 \end{bmatrix}$$

Once again the local forcing function is

$$\mathbf{f^e} = q\begin{bmatrix} L + x_e \\ L + x_{e+1} \\ L + x_{e+2} \end{bmatrix}$$

So the local forcing vector is

$$\mathbf{F^e} = \mathbf{M^e f^e} = q\frac{h^e}{30}\begin{bmatrix} 4x_e + 2x_{e+1} - x_{e+2} + 5L \\ 2x_e + 16x_{e+1} + 2x_{e+2} + 20L \\ -x_e + 2x_{e+1} + 4x_{e+2} + 5L \end{bmatrix}$$

The boundary terms are identical to before. Putting it all together we have

$$\mathbf{K}^\kappa + \bar{\mathbf{K}} = \frac{EA_0}{12L}\left(\begin{bmatrix} 62 + \frac{12Lk}{EA_0} & -72 & 10 & 0 & 0 & 0 & 0 \\ -72 & 160 & -88 & 0 & 0 & 0 & 0 \\ 10 & -88 & 140 & -72 & 10 & 0 & 0 \\ 0 & 0 & -72 & 160 & -88 & 0 & 0 \\ 0 & 0 & 10 & -88 & 140 & -72 & 10 \\ 0 & 0 & 0 & 0 & -72 & 160 & -88 \\ 0 & 0 & 0 & 0 & 10 & -88 & 78 \end{bmatrix} + 4\begin{bmatrix} 7 & -8 & 1 & 0 & 0 & 0 & 0 \\ -8 & 16 & -8 & 0 & 0 & 0 & 0 \\ 1 & -8 & 21 & -16 & 2 & 0 & 0 \\ 0 & 0 & -16 & 32 & -16 & 0 & 0 \\ 0 & 0 & 2 & -16 & 35 & -24 & 3 \\ 0 & 0 & 0 & 0 & -24 & 48 & -24 \\ 0 & 0 & 0 & 0 & 3 & -24 & 21 \end{bmatrix}\right)$$

$$\mathbf{F}^\kappa + \bar{\mathbf{F}} = q\frac{h^e}{30}\begin{bmatrix} 4x_1 + 2x_2 - x_3 + 5L \\ 2x_1 + 16x_2 + 2x_3 + 20L \\ -x_1 + 2x_2 + 8x_3 + 2x_4 - x_5 + 10L \\ 2x_3 + 16x_4 + 2x_5 + 20L \\ -x_3 + 2x_4 + 8x_5 + 2x_6 - x_7 + 10L \\ 2x_5 + 16x_6 + 2x_7 + 20L \\ -1x_5 + 2x_+ 4x_7 + 5L + \frac{30F}{qh^e} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \\ a_6 \\ a_7 \end{bmatrix} = \begin{bmatrix} 0.2500 \\ 0.2537 \\ 0.2567 \\ 0.2590 \\ 0.2609 \\ 0.2623 \\ 0.2633 \end{bmatrix}$$

Finally, our choice of shape functions allows us to make the simplification

$$\tilde{u} = \phi_j a_j \Rightarrow \tilde{u} = \begin{cases} \phi_j^e a_{re+j-r} & x_{re+1-r} \leq x \leq x_{re+1} \end{cases}$$

Again, this applies $\forall e$; e is not in Einstein notation.

$$\phi_1^1(x) = \hat{\phi}_1(\xi^1(x)) = -\frac{1}{2}(\frac{2}{h^e}x - \frac{x_1+x_3}{h^e})\Big(1-(\frac{2}{h^e}x - \frac{x_1+x_3}{h^e})\Big) = 18x^2 - 9x + 1$$

$$\phi_2^1(x) = \hat{\phi}_2(\xi^1(x)) = \Big(1+(\frac{2}{h^e}x - \frac{x_1+x_3}{h^e})\Big)\Big(1-(\frac{2}{h^e}x - \frac{x_1+x_3}{h^e})\Big) = -36x^2 + 12x$$

$$\phi_3^1(x) = \hat{\phi}_3(\xi^1(x)) = \frac{1}{2}(\frac{2}{h^e}x - \frac{x_1+x_3}{h^e})\Big(1+(\frac{2}{h^e}x - \frac{x_1+x_3}{h^e})\Big) = 18x^2 - 3x$$

$$\phi_1^2(x) = \hat{\phi}_1(\xi^2(x)) = -\frac{1}{2}(\frac{2}{h^e}x - \frac{x_3+x_5}{h^e})\Big(1-(\frac{2}{h^e}x - \frac{x_3+x_5}{h^e})\Big) = 18x^2 - 21x + 6$$

$$\phi_2^2(x) = \hat{\phi}_2(\xi^2(x)) = \Big(1+(\frac{2}{h^e}x - \frac{x_3+x_5}{h^e})\Big)\Big(1-(\frac{2}{h^e}x - \frac{x_3+x_5}{h^e})\Big) = -36x^2 + 36x - 8$$

$$\phi_3^2(x) = \hat{\phi}_3(\xi^2(x)) = \frac{1}{2}(\frac{2}{h^e}x - \frac{x_3+x_5}{h^e})\Big(1+(\frac{2}{h^e}x - \frac{x_3+x_5}{h^e})\Big) = 18x^2 - 15x + 3$$

$$\phi_1^3(x) = \hat{\phi}_1(\xi^3(x)) = -\frac{1}{2}(\frac{2}{h^e}x - \frac{x_5+x_7}{h^e})\Big(1-(\frac{2}{h^e}x - \frac{x_5+x_7}{h^e})\Big) = 18x^2 - 33x + 15$$

$$\phi_2^3(x) = \hat{\phi}_2(\xi^3(x)) = \Big(1+(\frac{2}{h^e}x - \frac{x_5+x_7}{h^e})\Big)\Big(1-(\frac{2}{h^e}x - \frac{x_5+x_7}{h^e})\Big) = -36x^2 + 60x - 24$$

$$\phi_3^3(x) = \hat{\phi}_3(\xi^3(x)) = \frac{1}{2}(\frac{2}{h^e}x - \frac{x_5+x_7}{h^e})\Big(1+(\frac{2}{h^e}x - \frac{x_5+x_7}{h^e})\Big) = 18x^2 - 27x + 10$$

$$\Rightarrow \tilde{u} = \begin{cases} 0.2500(18x^2 - 9x + 1) + 0.2535(-36x^2 + 12x) + 0.2567(18x^2 - 3x) & 0 \leq x \leq \frac{1}{3} \\ 0.2567(18x^2 - 21x + 6) + 0.2590(-36x^2 + 36x - 8) + 0.2609(18x^2 - 15x + 3) & \frac{1}{3} \leq x \leq \frac{2}{3} \\ 0.2609(18x^2 - 33x + 15) + 0.2623(-36x^2 + 60x - 24) + 0.2633(18x^2 - 27x + 10) & \frac{2}{3} \leq x \leq 1 \end{cases}$$

To find the flux we differentiate the solution:

$$\tilde{q} = \frac{d\tilde{u}}{dx} = \begin{cases} \frac{d\phi_j^e(x)}{dx} a_{re+j-r} & x_{re+1-r} \leq x \leq x_{re+1} \end{cases}$$

$$= \begin{cases} B_j(\xi^e(x))\frac{2}{h^e} a_{re+j-r} & x_{re+1-r} \leq x \leq x_{re+1} \end{cases}$$

$$\Rightarrow \tilde{q} = \begin{cases} 0.2500(6x - \frac{3}{2}) + 0.2535(-12x - 2) + 0.2567(6x - \frac{1}{2}) & 0 \leq x \leq \frac{1}{3} \\ 0.2567(6x - \frac{7}{2}) + 0.2590(-12x - 6) + 0.2609(6x - \frac{5}{2}) & \frac{1}{3} \leq x \leq \frac{2}{3} \\ 0.2609(6x - \frac{11}{2}) + 0.2623(-12x - 10) + 0.2633(6x - \frac{9}{2}) & \frac{2}{3} \leq x \leq 1 \end{cases}$$

1.4

To solve the problem with $n = 2$ elements of order $r = 3$ we introduce the identical nodal points and transformation as before. For $r = 3$:

$$\begin{cases} 16\hat{\phi}_1(\xi^e) = -(3\xi_e + 1)(3\xi_e - 1)(\xi_e - 1) \\ 16\hat{\phi}_2(\xi^e) = 9(\xi_e + 1)(3\xi_e - 1)(\xi_e - 1) \\ 16\hat{\phi}_3(\xi^e) = -9(\xi_e + 1)(3\xi_e + 1)(\xi_e - 1) \\ 16\hat{\phi}_4(\xi^e) = (3\xi_e + 1)(3\xi_e - 1)(\xi_e + 1) \end{cases}$$

$$\begin{cases} 16\hat{B}_1(\xi^e) = -27\xi_e^2 + 18\xi_e + 1 \\ 16\hat{B}_2(\xi^e) = 81\xi_e^2 - 18\xi_e - 27 \\ 16\hat{B}_2(\xi^e) = -81\xi_e^2 - 18\xi_e + 27 \\ 16\hat{B}_1(\xi^e) = 27\xi_e^2 + 18\xi_e - 1 \end{cases}$$

Our integrand is order $(r-1)+(r-1)+1 = 2r-1 = 5$ so we need $2p-1 \geq 5 \Rightarrow \mathbb{N} \ni p \geq 3$ Gauss points for exact integral value.

$$\xi_1^e = -\sqrt{\frac{3}{5}};\ \xi_2^e = 0;\ \xi_3^e = \sqrt{\frac{3}{5}};\ w_1 = w_3 = \frac{5}{9};\ w_2 = \frac{8}{9}$$

$$K_{ij}^e = \sum_{k=1}^{3} w_k \hat{B}_i E A_0 \frac{1}{h^e n}(\xi^e + 2n + 2e - 1)\hat{B}_j$$

To calculate this matrix by hand would be tedious so a computer is used for the arithmetic. Again for the RHS we use mass matrix techniques with

$$\mathbf{M^e} = \frac{h}{1680}\begin{bmatrix} 128 & 99 & -36 & 19 \\ 99 & 648 & -81 & -36 \\ -36 & -81 & 648 & 99 \\ 19 & -36 & 99 & 128 \end{bmatrix}$$

Once again the local forcing function is

$$\mathbf{f^e} = q\begin{bmatrix} L + x_e \\ L + x_{e+1} \\ L + x_{e+2} \\ L + x_{e+3} \end{bmatrix}$$

So the local forcing vector is

$$\mathbf{F^e} = \mathbf{M^e f^e}$$

The boundary terms are identical to before. Putting it all together we have

$$(\mathbf{K}^\kappa + \bar{\mathbf{K}})\mathbf{a} = \mathbf{F^g} + \bar{\mathbf{F}} \Rightarrow$$

$$\Rightarrow \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \\ a_6 \\ a_7 \end{bmatrix} = \begin{bmatrix} 0.2500 \\ 0.2537 \\ 0.2567 \\ 0.2590 \\ 0.2609 \\ 0.2623 \\ 0.2633 \end{bmatrix}$$

Finally, our choice of shape functions allows us to make the simplification

$$\tilde{u} = \phi_j a_j \Rightarrow \tilde{u} = \begin{cases} \phi_j^e a_{re+j-r} & x_{re+1-r} \leq x \leq x_{re+1} \end{cases}$$

Again, this applies $\forall e$; e is not in Einstein notation.

$$\phi_1^1(x) = \hat{\phi}_1(\xi^1(x)) = -(3(\frac{2}{h^e}x - \frac{x_1+x_4}{h^e})+1)(3(\frac{2}{h^e}x - \frac{x_1+x_4}{h^e})-1)((\frac{2}{h^e}x - \frac{x_1+x_4}{h^e})-1) = -(2x-1)(3x-1)(6x-1)$$

$$\phi_2^1(x) = \hat{\phi}_2(\xi^1(x)) = 9((\frac{2}{h^e}x - \frac{x_1+x_4}{h^e})+1)(3(\frac{2}{h^e}x - \frac{x_1+x_4}{h^e})-1)((\frac{2}{h^e}x - \frac{x_1+x_4}{h^e})-1) = 18x(2x-1)(3x-1)$$

$$\phi_3^1(x) = \hat{\phi}_3(\xi^1(x)) = -9((\frac{2}{h^e}x - \frac{x_1+x_4}{h^e})+1)(3(\frac{2}{h^e}x - \frac{x_1+x_4}{h^e})+1)((\frac{2}{h^e}x - \frac{x_1+x_4}{h^e})-1) = -9x(2x-1)(6x-1)$$

$$\phi_4^1(x) = \hat{\phi}_4(\xi^1(x)) = (3(\frac{2}{h^e}x - \frac{x_1+x_4}{h^e})+1)(3(\frac{2}{h^e}x - \frac{x_1+x_4}{h^e})-1)((\frac{2}{h^e}x - \frac{x_1+x_4}{h^e})+1) = 32x(3x-1)(6x-1)$$

$$\phi_1^2(x) = \hat{\phi}_1(\xi^2(x)) = -(3(\frac{2}{h^e}x - \frac{x_4+x_7}{h^e})+1)(3(\frac{2}{h^e}x - \frac{x_4+x_7}{h^e})-1)((\frac{2}{h^e}x - \frac{x_4+x_7}{h^e})-1) = -2(x-1)(3x-2)(6x-5)$$

$$\phi_2^2(x) = \hat{\phi}_2(\xi^2(x)) = 9((\frac{2}{h^e}x - \frac{x_4+x_7}{h^e})+1)(3(\frac{2}{h^e}x - \frac{x_4+x_7}{h^e})-1)((\frac{2}{h^e}x - \frac{x_4+x_7}{h^e})-1) = 9(x-1)(2x-1)(6x-5)$$

$$\phi_3^2(x) = \hat{\phi}_3(\xi^2(x)) = -9((\frac{2}{h^e}x - \frac{x_4+x_7}{h^e})+1)(3(\frac{2}{h^e}x - \frac{x_4+x_7}{h^e})+1)((\frac{2}{h^e}x - \frac{x_4+x_7}{h^e})-1) = -18(x-1)(2x-1)(3x-2)$$

$$\phi_4^2(x) = \hat{\phi}_4(\xi^2(x)) = (3(\frac{2}{h^e}x - \frac{x_4+x_7}{h^e})+1)(3(\frac{2}{h^e}x - \frac{x_4+x_7}{h^e})-1)((\frac{2}{h^e}x - \frac{x_4+x_7}{h^e})+1) = 16(2x-1)(3x-2)(6x-5)$$

$$\Rightarrow \tilde{u} = \begin{cases} u_1 & 0 \leq x \leq \frac{1}{2} \\ u_2 & \frac{1}{2} \leq x \leq 1 \end{cases}$$

$$u_1 = 0.2500(-(2x-1)(3x-1)(6x-1)) + 0.2537 \cdot 18x(2x-1)(3x-1) + 0.2567(-9x(2x-1)(6x-1)) \\ +0.2590(32x(3x-1)(6x-1))$$

$$u_2 = 0.2590(-2(x-1)(3x-2)(6x-5)) + 0.2609 \cdot 9(x-1)(2x-1)(6x-5) + 0.2623(-18(x-1)(2x-1)(3x-2)) \\ +0.2633(16(2x-1)(3x-2)(6x-5))$$

To find the flux we differentiate the solution:

$$\tilde{q} = \frac{d\tilde{u}}{dx} = \Big\{ \frac{d\phi_j^e(x)}{dx} a_{re+j-r} \quad x_{re+1-r} \leq x \leq x_{re+1}$$

$$= \Big\{ B_j(\xi^e(x)) \frac{2}{h^e} a_{re+j-r} \quad x_{re+1-r} \leq x \leq x_{re+1}$$

$$\Rightarrow \tilde{q} = \begin{cases} q_1 & 0 \leq x \leq \frac{1}{2} \\ q_2 & \frac{1}{2} \leq x \leq 1 \end{cases}$$

$$q_1 = 0.2500(-11+72x-108x^2) + 0.2537(18(1-10x+18x^2)) + 0.2567(-9(1-16x+36x^2)) + 0.2590(32(1-18x+54x^2))$$

$$q_2 = 0.2590(-2(37-90x+54x^2)) + 0.2609(9(21-56x+36x^2)) + 0.2623(-18(9-26x+18x^2)) + 0.2633(16(47-144x+108x^2))$$

1.5

MATLAB code attached.

1.6

1.7

See appendix for figures.

1.8

To aid the calculations of L_2 and H_1 the Gaussian quadrature was done "on paper:"

$$L_2 = \sqrt{\int_\Omega (u-\tilde{u})^2 dx} = \sqrt{\sum_{e=1}^{n} \int_{x_{er-r+1}}^{x_{er+1}} (u(x) - \tilde{u}(x))^2 dx}$$

$$= \sqrt{\sum_{e=1}^{n} \int_{-1}^{1} \Big(u(x(\xi^e)) - \tilde{u}(x(\xi^e)) \Big)^2 |J^e| d\xi^e}$$

Using $\tilde{u} = \phi_j a_j \Rightarrow \tilde{u} = \Big\{ \phi_j^e a_{er+j-r} \quad x_{er+1-r} \leq x \leq x_{er+1}$:

$$\tilde{u}(x(\xi^e)) = a_{er+j-r}\phi_j^e(x(\xi^e)) = a_{er+j-r}\hat{\phi}_j(\xi^e)$$

$$L_2 = \sqrt{\sum_{e=1}^{n} \int_{-1}^{1} \Big(u(x(\xi^e)) - a_{er+j-r}\hat{\phi}_j(\xi^e) \Big)^2 |J^e| d\xi^e}$$

We can make the simplification because we are only evaluating this term within the bounds of the integral, which are the bounds of the element. As for H_1:

$$H_1 = \sqrt{\int_\Omega \left((u-\tilde{u})^2 + (\frac{du}{dx} - \frac{d\tilde{u}}{dx})^2\right) dx} = \sqrt{\sum_{e=1}^{n} \int_{x_{re+1-r}}^{x_{re+1}} \left(\left(u(x) - \tilde{u}(x)\right)^2 + (\frac{du(x)}{dx} - \frac{d\tilde{u}(x)}{dx})^2\right) dx}$$

$$= \sqrt{\sum_{e=1}^{n} \int_{-1}^{1} \left[\left(u(x(\xi^e)) - \tilde{u}(x(\xi^e))\right)^2 + \left(\frac{du(x)}{dx}\bigg|_{x(\xi^e)} - \frac{d\tilde{u}(x)}{dx}\bigg|_{x(\xi^e)}\right)^2\right] |J^e| d\xi^e}$$

Again, since we are only evaluating these two $\tilde{u}$-terms within the bounds of the integral, we can simplify to

$$H_1 = \sqrt{\sum_{e=1}^{n} \int_{-1}^{1} \left[\left(u(x(\xi^e)) - a_{er+j-r}\hat{\phi}_j(\xi^e)\right)^2 + \left(\frac{du(x)}{dx}\bigg|_{x(\xi^e)} - a_{er+j-r}B_j(\xi^e)\frac{2}{h^e}\right)^2\right] \frac{h^e}{2} d\xi^e}$$

We have not explored Gauss quadrature with logarithmic integrands (such as in the exact solution here) so we will simply use a relatively large number of Gauss points, $p = 4$.

2

2.1

We are asked to calculate certain displacements and forces for $k \to \infty$. This means the bottom rod, placed between two rigid nodes, becomes redundant, and we are left with a two-element problem. We could solve this, but we will instead continue directly to part 2 of the question, where $k = k_0$. After this, to answer the current part of the question, we will simply observe the behavior of the system as $k_0 \to \infty$ and check that it makes sense.

2.2

To obtain somewhat realistic results the values we will use for all rods for the problem are:

$$E = 10^{10}[Pa];\ A = 0.0001[m^2];\ P = 10^4[N];\ k_0 = 10^5[N/m];\ L = 1[m];\ \sigma_y = 276 \cdot 10^6[Pa]$$

These were gathered from online resources to somewhat resemble a setup with aluminum rods, industrial springs, and a relatively significant force.

(a)

We will refer to the right, left, and top nodes with subscripts R, L, T respectively, and to the left, right, and bottom elements with subscripts $1, 2, 3$ respectively. To incorporate the springs into truss element formulation we must first assume that nodal displacements perpendicular to the direction of any springs are small such that their effect on the deviation from free length of said springs is negligible. This way we can introduce constraints on the displacement $[u, v]$ and force $[P, Q]$ at relevant nodes according to Hooke's law:

$$P_L = -k_0 u_L$$
$$Q_L = -k_0 v_L$$
$$P_R = -k_0 u_R$$
$$Q_R = -k_0 v_R$$

At this point we can also introduce the boundary conditions for the top node. The triangle formed by the truss is equilateral and therefore the acute angle of the initial orientation of the left bar with respect to the horizontal is $\frac{\pi}{3}[rad]$. We will also denote the reaction force, necessarily perpendicular to the roller joint, as R:

$$P_T = R\sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}R$$
$$Q_T = P - R\cos\frac{\pi}{3} = P - \frac{R}{2}$$
$$\tan\frac{\pi}{3} = \frac{v_T}{u_T} \Rightarrow \sqrt{3}u_T = v_T$$

(b)

Essential boundary conditions are defined as BC's applied to the displacement of the truss. We require four essential BC's to match the four degrees of freedom (translation and rotation) of the truss. We have five essential BC's: one from each mixed (robin) BC from the springs, and one from the roller joint. Therefore, the problem is well-defined.

(c)

In order to determine transformation matrices, we must once again assume that nodal displacements are small such that their effect on rod angles with respect to the horizontal are negligible. We can then define these rod angles, the local stiffness matrices, local forcing vectors and transformation matrices:

$$\alpha^1 = \frac{\pi}{3}[rad];\ \alpha^2 = -\frac{\pi}{3}[rad];\ \alpha^3 = 0$$

$$\hat{\mathbf{K}}^{\mathbf{e}} = \frac{EA}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix};\ \mathbf{T}^{\mathbf{e}} = \begin{bmatrix} \cos\alpha^e & \sin\alpha^e & 0 & 0 \\ 0 & 0 & \cos\alpha^e & \sin\alpha^e \end{bmatrix};\ \mathbf{K}^{\mathbf{e}} = (\mathbf{T}^{\mathbf{e}})^T\hat{\mathbf{K}}^{\mathbf{e}}(\mathbf{T}^{\mathbf{e}});\ \mathbf{Ka} = \mathbf{F}$$

To create the global stiffness matrix we combine equations in the system which refer to the same node in an operation similar to the A-sum operation defined in tutorials. To create the external forces vector we insert the boundary conditions:

$$\mathbf{K} = \begin{bmatrix} K^1_{1,1} + K^3_{3,3} & K^1_{1,2} + K^3_{3,4} & K^1_{1,3} & K^1_{1,4} & K^3_{3,1} & K^3_{3,2} \\ K^1_{2,1} + K^3_{4,3} & K^1_{2,2} + K^3_{4,4} & K^1_{2,3} & K^1_{2,4} & K^3_{4,1} & K^3_{4,2} \\ K^1_{3,1} & K^1_{3,2} & K^1_{3,3} + K^2_{1,1} & K^1_{3,4} + K^2_{1,2} & K^2_{1,3} & K^2_{1,4} \\ K^1_{4,1} & K^1_{4,2} & K^1_{4,3} + K^2_{2,1} & K^1_{4,4} + K^2_{2,2} & K^2_{2,3} & K^2_{2,4} \\ K^3_{1,3} & K^3_{1,4} & K^2_{3,1} & K^2_{3,2} & K^2_{3,3} + K^3_{1,1} & K^2_{3,4} + K^3_{1,2} \\ K^3_{2,3} & K^3_{2,4} & K^2_{4,1} & K^2_{4,2} & K^2_{4,3} + K^3_{2,1} & K^2_{4,4} + K^3_{2,2} \end{bmatrix}$$

$$\mathbf{a} = \begin{bmatrix} u_L \\ v_L \\ u_T \\ v_T \\ u_R \\ v_R \end{bmatrix};\ \mathbf{F} = \begin{bmatrix} -k_0 u_L \\ -k_0 v_L \\ \frac{\sqrt{3}}{2}R \\ P - \frac{R}{2} \\ -k_0 u_R \\ -k_0 v_R \end{bmatrix}$$

We don't know the normal force at the top node. We can eliminate it from the fourth row using the third row, and then replace the third row with the final unused boundary condition on the top node. The unused third row equation can simply be used later to solve for this unknown force. We rewrite the system to:

$$\begin{bmatrix} K^1_{1,1} + K^3_{3,3} + k_0 & K^1_{1,2} + K^3_{3,4} & K^1_{1,3} & K^1_{1,4} & K^3_{3,1} & K^3_{3,2} \\ K^1_{2,1} + K^3_{4,3} & K^1_{2,2} + K^3_{4,4} + k_0 & K^1_{2,3} & K^1_{2,4} & K^3_{4,1} & K^3_{4,2} \\ 0 & 0 & \sqrt{3} & -1 & 0 & 0 \\ \textit{fourth row} & & & & & \\ K^3_{1,3} & K^3_{1,4} & K^2_{3,1} & K^2_{3,2} & K^2_{3,3} + K^3_{1,1} + k_0 & K^2_{3,4} + K^3_{1,2} \\ K^3_{2,3} & K^3_{2,4} & K^2_{4,1} & K^2_{4,2} & K^2_{4,3} + K^3_{2,1} & K^2_{4,4} + K^3_{2,2} + k_0 \end{bmatrix} \begin{bmatrix} u_L \\ v_L \\ u_T \\ v_T \\ u_R \\ v_R \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ P \\ 0 \\ 0 \end{bmatrix}$$

Abusive notation is only used here temporarily because the matrix doesn't fit on the page otherwise.

$$(\textit{fourth row})^T = \begin{bmatrix} K^1_{4,1} + \frac{K^1_{3,1}}{\sqrt{3}} \\ K^1_{4,2} + \frac{K^1_{3,2}}{\sqrt{3}} \\ K^1_{4,3} + K^2_{2,1} + \frac{K^1_{3,3} + K^2_{1,1}}{\sqrt{3}} \\ K^1_{4,4} + K^2_{2,2} + \frac{K^1_{3,4} + K^2_{1,2}}{\sqrt{3}} \\ K^2_{2,3} + \frac{K^2_{1,3}}{\sqrt{3}} \\ K^2_{2,4} + \frac{K^2_{1,4}}{\sqrt{3}} \end{bmatrix}$$

We can solve this system in MATLAB to get

$$\mathbf{a} = \begin{bmatrix} 0.0082 \\ 0.0580 \\ 0.0305 \\ 0.0529 \\ 0.0057 \\ 0.0340 \end{bmatrix}$$

Therefore the displacement vector of the top hinge is $\begin{bmatrix} 0.0305 & 0.0529 \end{bmatrix} [m]$.
To answer part 1 of the question: We can solve the system again in MATLAB for a very large k_0, checking that all nodal displacements for the left and right nodes vanish. This way we get that the displacement vector of the top hinge is $\begin{bmatrix} 0.0035 & 0.0060 \end{bmatrix} [m]$.
To calculate the force which will cause the bar to yield we will first identify the critical rod in the truss: the left rod. As the top node is confined by the roller joint, it is easy to see that the left rod elongates more than the right rod as force increases. The left node doesn't move so the extension of the left rod is just the magnitude of the displacement of the top node in the rod's direction. Therefore we are looking for a force P such that

$$\frac{\sigma_y}{E} = u_T \cos\frac{\pi}{3} + v_T \sin\frac{\pi}{3}$$

We can repeatedly solve the system for different values of P until this equation holds to conclude that under this condition, $P = 39837[N]$.

Appendix

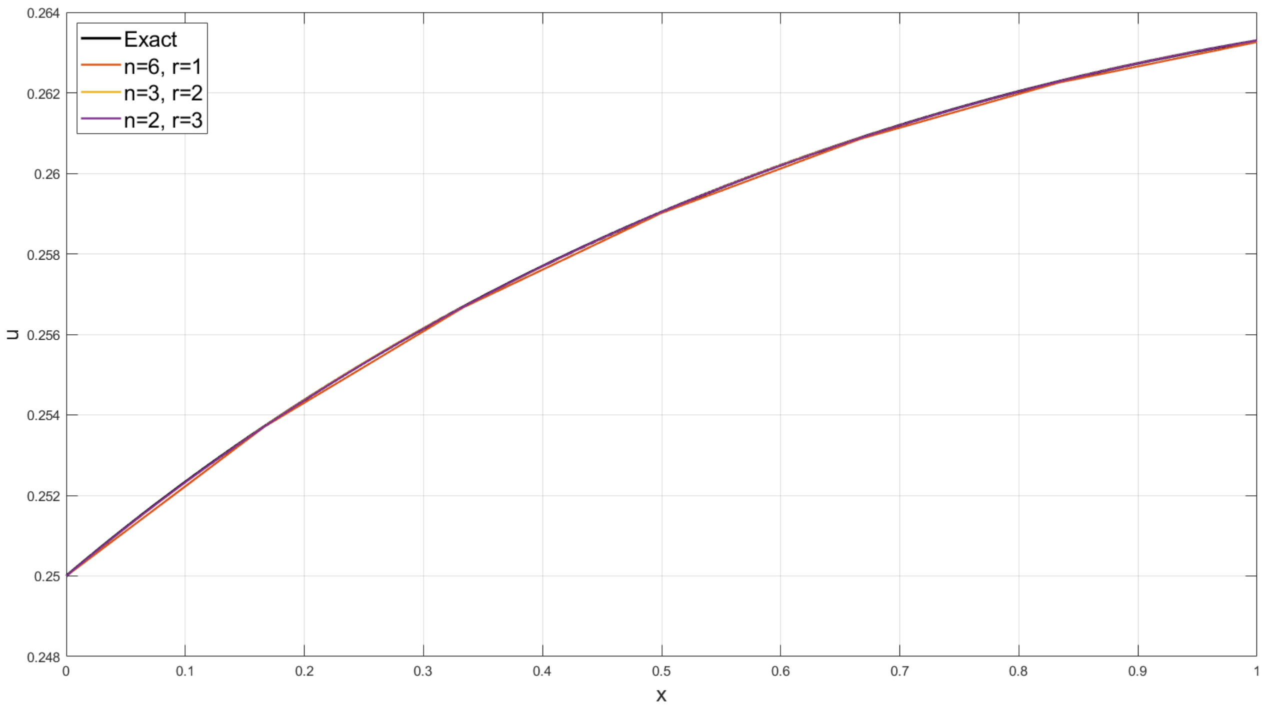

Figure 1: Q1.6: Exact Solution vs. Numerical Solution Using 6 Linear, 3 Quadratic, 2 Cubic Elements

The resulting rates of convergence for each error norm are listed in the figures.

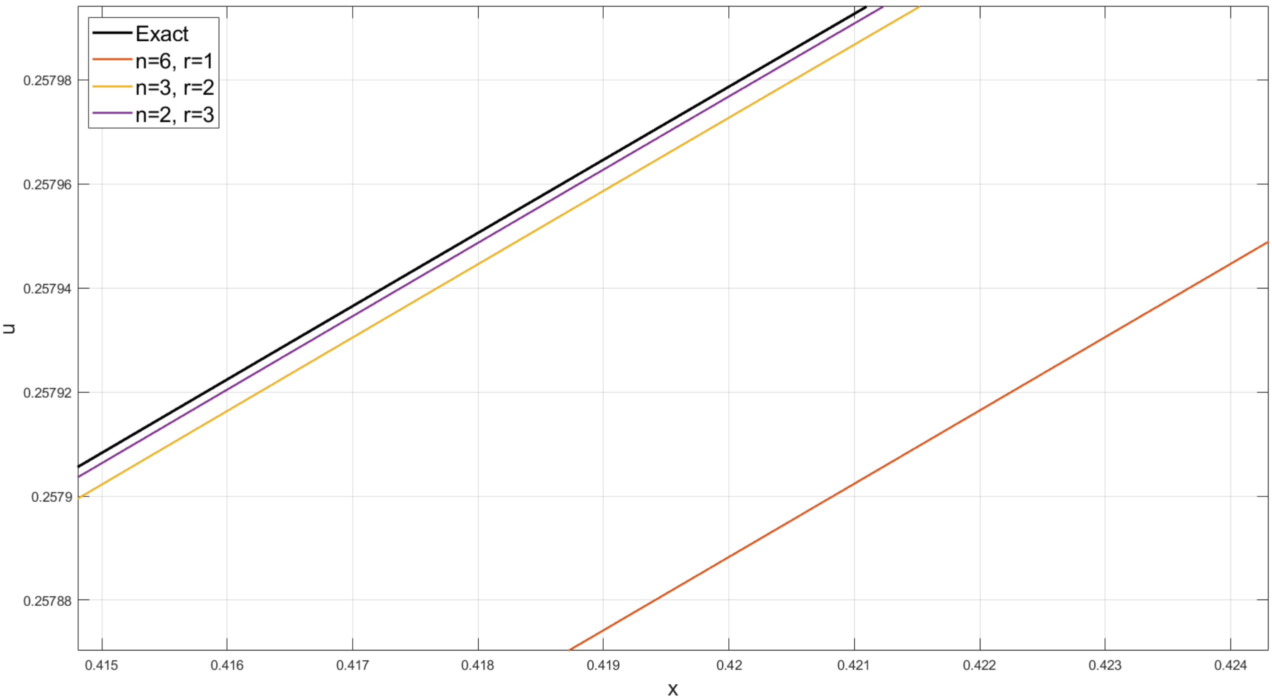

Figure 2: Q1.6: Above graph, zoomed in for visibility

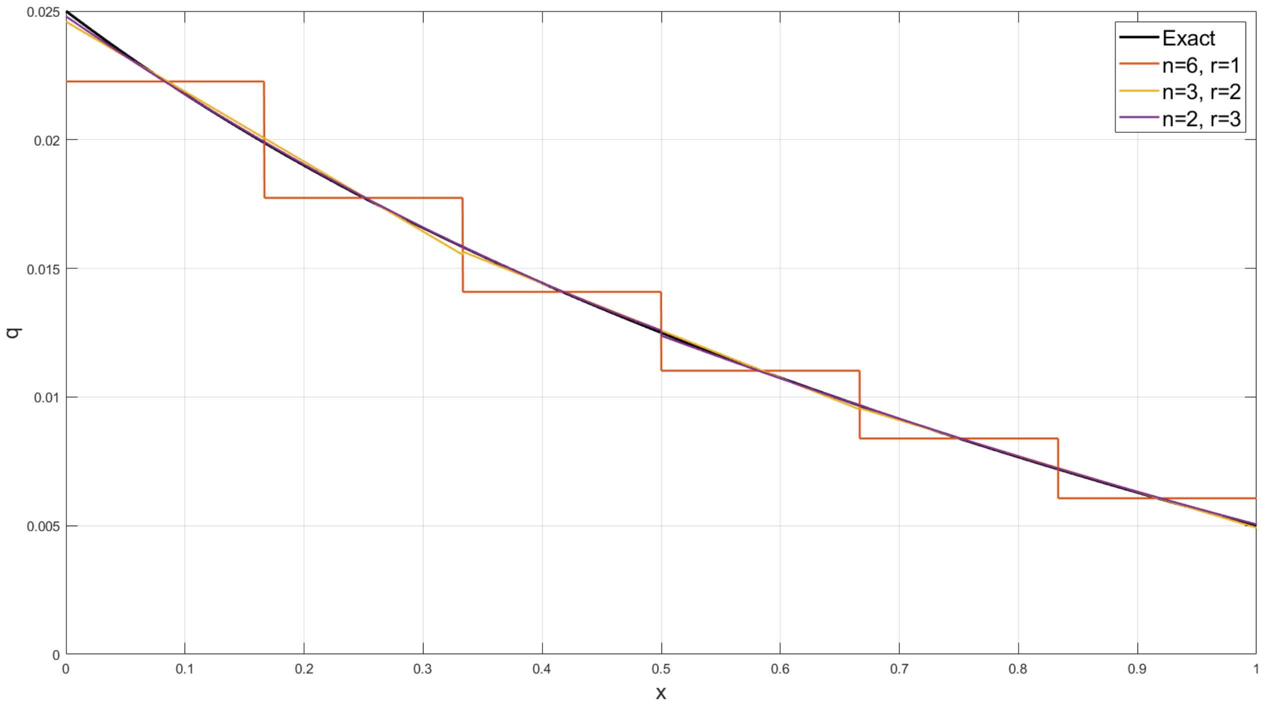

Figure 3: Q1.7: Exact Flux of Solution vs. Numerical Flux Using 6 Linear, 3 Quadratic, 2 Cubic Elements

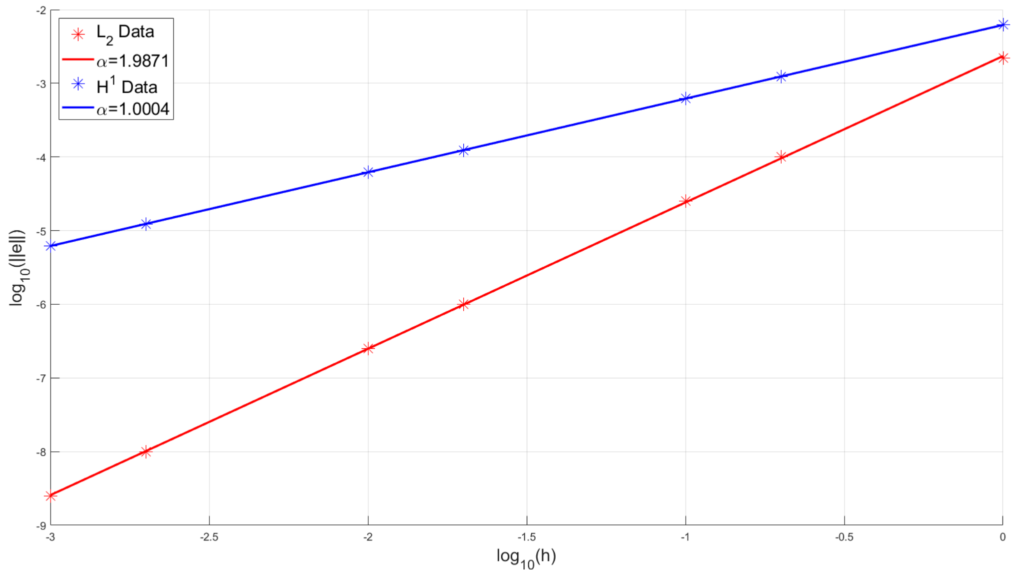

Figure 4: Q1.8: Logarithmic Error Norms vs. Logarithmic Linear Element Size h

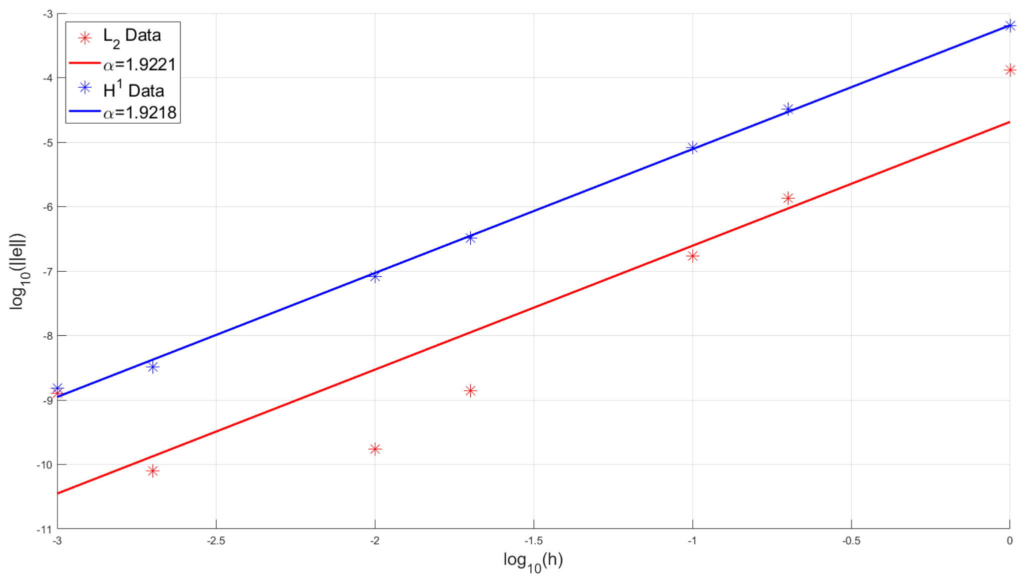

Figure 5: Q1.8: Logarithmic Error Norms vs. Logarithmic Quadratic Element Size h

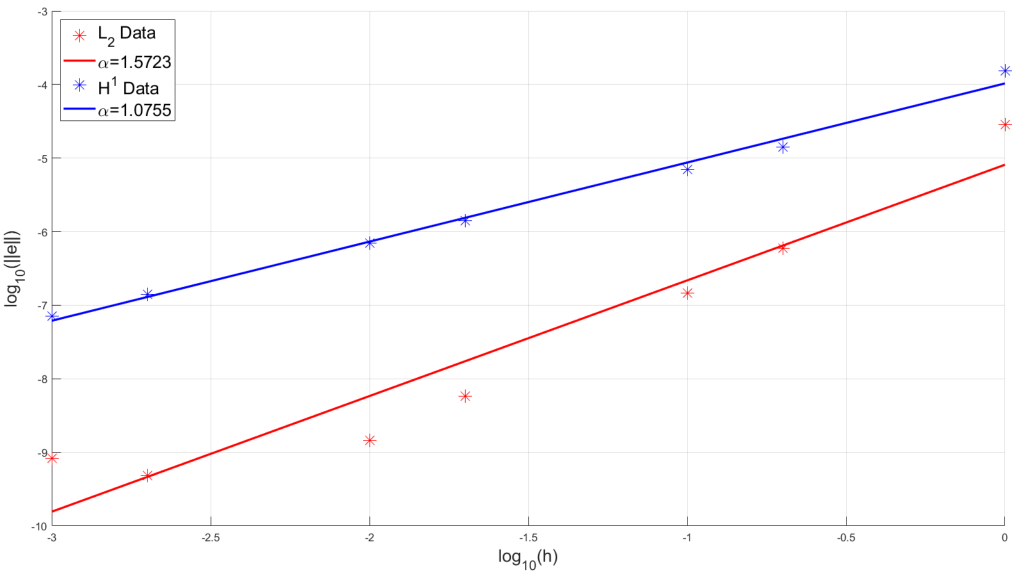

Figure 6: Q1.8: Logarithmic Error Norms vs. Logarithmic Cubic Element Size h